When Fault-Aware World-Model Prediction Helps Control: A Six-Stage Diagnosis under Joint Locking

Anonymous Authors

***Abstract*—A diagnosed joint lock changes a robot's feasible state manifold, but a lower world-model prediction error need not identify better contact actions. We study few-shot fault adaptation through a six-stage protocol: constraint satisfaction, reachability, contact establishment, contact-response prediction, action ranking, and realized candidate outcome. The frozen model family combines analytic lock projection, a learned fault residual, and paired counterfactual decision training. Under one common five-DoF pushing protocol, three seeds, 400 action groups per seed, 128 candidates per group, and 50-step rollouts, a same-capacity global fault residual reduces top-1 action regret by 19.76% over the nominal world model (10.66–27.86%; 3/3 seeds) and terminal candidate error by 4.04% (3/3), although contact-response RMSE worsens by 270.04%. Removing analytic projection produces 0.077–0.153 rad lock-position drift and 0.384–0.660 rad/s velocity violation; projection makes both exactly zero in 3/3 seeds. A selective intervention model does not improve aggregate regret, terminal error, or success over the same-capacity global residual, so we reject selective-structure attribution. The result is a reproducible diagnosis rather than a universal recovery claim: hard physical correctness and useful action ranking are distinct from average predictive accuracy.**

I. Introduction

A robot that loses one joint does not merely acquire a new continuous dynamics parameter. Its feasible state manifold changes discretely. Diagnosis may provide the locked joint and angle, yet contact outcomes remain uncertain because damping, actuation delay, deadband, friction, and payload are only partially observed. Re-fitting a complete dynamics model after failure is data-intensive, while unconstrained adaptation can violate the diagnosed lock or degrade unaffected coordinates.

Prior damage-recovery methods adapt policies or behavior repertoires [1], [2]; rapid adaptation infers latent physical properties from recent transitions [3], [4]; and learned pushing models capture uncertain contact dynamics [5], [6], [7]. Their combination leaves two questions that are often conflated: does a fault-aware predictor obey the diagnosed feasible manifold, and does its numerical prediction improvement preserve the ordering of actions needed by a planner?

Our starting point is a disclosed sequence of failures. Hard masking produced feasible-looking rollouts without establishing contact; lower object RMSE did not consistently reduce task error; and a selective intervention branch did not outperform a same-capacity global residual. These failures motivate evaluation at each causal gate rather than another aggregate state-error table.

The contributions are:

TABLE I
MECHANISM-LEVEL SCOPE RELATIVE TO REPRESENTATIVE ADAPTATION WORK. "EXACT" DENOTES CONSTRUCTION RATHER THAN A LEARNED PENALTY.

Method	Robot topology	Object physics	Exact state isolation
RMA [3]	continuous latent	limited	no
TuneNet [4]	no	identified	no
AdaptiGraph [11]	no	few-shot	no
PIN-WM [12]	no	few-shot	no
SI-IPWM	held-out lock	few-shot	yes

- a six-stage evaluation that prevents lock feasibility, contact, response prediction, action ranking, and task outcome from being collapsed into one RMSE;
- an analytic diagnosis-conditioned projection whose exact constraint benefit is measured by a three-seed removal ablation;
- a controlled comparison of nominal, projected carrier, same-capacity global residual, full-state and selective residual, and decision-loss variants under identical candidate data and planning budget; and
- an auditable positive/negative result: fault adaptation consistently reduces action regret while response RMSE degrades, whereas selective publication fails its same-capacity attribution test.

II. Related Work

Damage adaptation. Repertoire search [1], fault-aware reinforcement learning [2], latent extrinsics [3], simulator tuning [4], and meta-learned adaptive control [8] address recovery under changed dynamics. We instead assume a diagnosed discrete lock and adapt a predictive interaction model while enforcing the resulting constraint.

Contact dynamics. Learned forward models support visual and model-based planning [5], [9]; few-sample GP models [6], online identification [10], and equivariant pushing models [7] improve data efficiency. Recent systems infer physical properties in graph or physics-informed world models [11], [12] or combine residual physics with active planning [13]. Interaction networks encode relational structure [14], [15]. Our focus is complementary: isolating an object-specific learned correction from robot coordinates whose feasible manifold is known after diagnosis.

Table I is a scope comparison, not a global-superiority claim. The closest methods model richer object families or active data acquisition; our narrower novelty is the interface between diagnosed robot topology and an object correction that cannot rewrite published robot state.

III. Problem Formulation

The state is $s_t = [r_t, o_t]$, where $r_t = [q_t, \dot{q}_t] \in \mathbb{R}^{10}$ and planar object position/velocity $o_t \in \mathbb{R}^4$; action $a_t \in \mathbb{R}^5$. Diagnosis $d = (m, \bar{q})$ supplies a binary lock mask and lock angle. Unknown physical context $z \in \mathbb{R}^8$ is inferred from $K = 25$ safe transitions. Joint D3 is absent from training and validation and is combined at test time with seven continuous-physics conditions.

We evaluate autonomous rollouts at $H \in \{10, 25, 50\}$. Object RMSE is the primary metric. Free-joint RMSE excludes locked coordinates; pusher-position RMSE evaluates forward-kinematic propagation; and lock violation measures deviation from $q^j = \bar{q}^j, \dot{q}^j = 0$. Calibration and evaluation trajectories are disjoint.

IV. Method

A. Projection and base dynamics

For each joint,

$$\Pi_d(q)_j = (1 - m_j)q_j + m_j\bar{q}_j, \tag{1}$$
$$\Pi_d(\dot{q})_j = (1 - m_j)\dot{q}_j, \qquad \Pi_d(a)_j = (1 - m_j)a_j. \tag{2}$$

Projection is applied before and after prediction. A contact-aware recurrent base predicts robot and object state; forward kinematics supplies pusher geometry. The intervention branch adds geometry and low-rank latent object residuals modulated by observable context. Base checkpoints and the robot block are frozen before intervention training.

Proposition 1 (analytic feasibility). Π_d is idempotent. After every projected transition, each locked joint satisfies $q^j = \bar{q}^j$, $\dot{q}^j = 0$, and $a^j = 0$, independently of learned parameters and rollout depth.

Proof. Substituting the binary mask twice leaves locked entries at $(\bar{q}^j, 0, 0)$ and leaves free entries unchanged, hence $\Pi_d \circ \Pi_d = \Pi_d$. Applying this identity after each transition gives the finite-depth statement by induction. □

B. Physical-support routing

An amortized encoder estimates context from the K25 transition sequence. A centered modulation $g(z) = h(z) - h(0)$ gives exact recovery at zero context. A development audit showed that topology membership alone activates harmful corrections in nominal-like physics. We therefore route intervention only if $\|z\|_2 \geq 1.2$, a threshold frozen after seed 27; otherwise the method is the carrier exactly. This rule activates on high damping, mixed composition, and mixed unseen physics in the reported audit and falls back on nominal, weak motor, delay, and noisy-deadband conditions.

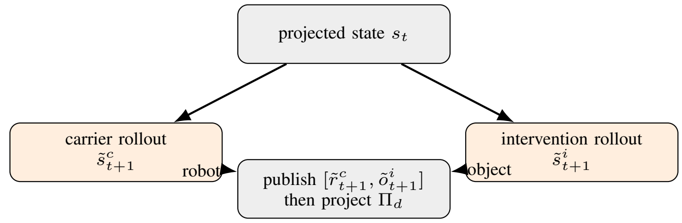

Fig. 1. State isolation. Both models retain private recurrent and predicted physical states. Only the published state mixes carrier robot coordinates with intervention object coordinates.

C. State-isolated rollout

Let F_c be the no-intervention carrier and F_i the full intervention model. Each maintains private hidden and physical states:

$$(\tilde{s}^c_{t+1}, h^c_{t+1}) = F_c(\tilde{s}^c_t, a_t, d, h^c_t), \tag{3}$$
$$(\tilde{s}^i_{t+1}, h^i_{t+1}) = F_i(\tilde{s}^i_t, a_t, d, z, h^i_t). \tag{4}$$

The externally visible prediction is

$$\hat{s}_{t+1} = \Pi_d([\tilde{r}^c_{t+1}, \tilde{o}^i_{t+1}]). \tag{5}$$

Crucially, $\hat{s}_{t+1}$ is not fed back into either private dynamics path. Thus the intervention preserves the robot–object feedback it learned, while Eq. (5) makes published robot coordinates identical to the carrier for any action sequence. Pusher position, being a deterministic function of q, is also identical. Projection gives zero lock violation.

This guarantee is relative to the chosen carrier, not ground truth: state isolation prevents collateral prediction change but cannot correct a bad carrier. It also does not guarantee that planning with the improved object prediction produces better actions.

Proposition 2 (published-state isolation). For identical initial state, actions, diagnosis, and carrier parameters, the published robot trajectory of SI-IPWM equals the standalone carrier trajectory at every depth. Its pusher trajectory is therefore identical, and locked-coordinate violation is zero.

Proof. At depth zero both carrier states are identical. Assuming equality at depth t, both standalone and internal carriers apply the same deterministic F_c, action, diagnosis, hidden state, and projection, so their robot states and next hidden states agree. Equation (5) copies that robot state without modification except the same Π_d. Induction gives equality for all depths. Pusher pose is deterministic forward kinematics of q, and Π_d fixes locked position and velocity. □

Proposition 3 (conditional guarded fallback). Let A be proposal acceptance, S certificate validity, and J finite-horizon cost. If valid accepted proposals satisfy $J(\pi_i) - J(\pi_c) \leq L\epsilon_H$ and every false acceptance has excess cost at most C, exact carrier fallback gives

$$\mathbb{E}[J(\pi_g) - J(\pi_c)] \leq L\mathbb{E}[\epsilon_H \mathbf{1}_{A \cap S}] + C\Pr(A \cap S^c). \tag{6}$$

Rejection contributes zero because the published action is exactly π_c. This is a conditional decomposition, not a numerical safety claim: the present experiments do not yet establish a sufficiently small false-accept term.

D. Training objective and deployment

The fault residual is trained after freezing the shared base and robot block. For rollout depths $\mathcal{H}$, its prediction loss is

$$\mathcal{L}_{\text{pred}} = \sum_{H \in \mathcal{H}} w_H (\lambda_r \mathcal{L}_r^H + \lambda_o \mathcal{L}_o^H + \lambda_p \mathcal{L}_p^H) + \lambda_T \mathcal{L}_o^{50}. \quad (7)$$

For candidates $k = 1, \ldots, K$ sharing an initial state, diagnosis, and goal, let $\hat{c}_k$ be predicted terminal object-to-goal cost, c_k its realized counterpart, and $c^* = \min_k c_k$. We additionally optimize

$$\mathcal{L}_{\text{dec}} = \sum_k \frac{\exp(-\hat{c}_k/\tau)}{\sum_j \exp(-\hat{c}_j/\tau)} (c_k - c^*), \quad \mathcal{L} = \mathcal{L}_{\text{pred}} + \lambda_d \mathcal{L}_{\text{dec}}. \quad (8)$$

This changes no deployed module: at test time the world model rolls every candidate forward and selects minimum predicted terminal cost. The paired objective is evaluated separately from weight zero because it may trade average prediction error for ranking quality. Feasibility is absent from the loss because projection enforces the diagnosed lock by construction.

We compare two equal-budget residual supports. The global residual may correct all predicted state coordinates. The selective model keeps a private coupled trajectory but publishes robot coordinates from the carrier and object coordinates from the intervention branch. Both share projection, initialization, candidate data, epochs, and selection rule; this comparison is the attribution test rather than a claim based on parameter count alone.

The strict checkpoints contain 337,834 parameters for selective IPWM and 337,842 for the same-capacity global control. Both train for 40 epochs; epoch selection uses only the frozen D2/D4 development split. Full checkpoint loading requires exact key and shape equality, preventing untrained object or intervention heads from entering formal evaluation.

Formal 400-by-128-by-50 evaluation takes 114–333 s for selective IPWM and 133–150 s for the global residual across recorded runs. These timings mix CPU and GPU runs and therefore document resource cost rather than speed superiority. The GPU environment is an RTX 4060 Laptop GPU with PyTorch 2.11/CUDA 12.8. Historical training logs did not persist reliable end-to-end wall time, so no training-time estimate is reported.

V. Experimental Protocol

Experiments use MuJoCo planar pushing with the original five-joint serial-arm geometry and a four-dimensional planar object state. D2 and D4 locks form the frozen development distribution. Seeds 7/17/27 use separate model initializations and candidate archives. Formal evaluation contains 400 matched groups per seed, 128 action sequences per group, and 50 simulator steps per sequence. Within a group, every method receives exactly the same current state, goal, diagnosis, action candidates, physical contact labels, continuous minimum geometry distance, and realized terminal object state.

TABLE II

Strict three-seed results versus the nominal world model. Positive regret/error entries denote reductions; success is percentage points.

Method	Response RMSE	Spearman	Top-1 regret	Terminal error	Success
Global matched	-270.04%	$+0.0375$	$+19.76\%$	$+4.04\%$	$+1.58$
Selective IPWM	-247.90%	$+0.0378$	$+18.28\%$	$+3.73\%$	$+1.17$

The formal rows are nominal WM, analytic projection/carrier, same-capacity global residual, full-state residual, selective publication, decision weight zero, decision weight 10, and an oracle that selects by realized terminal cost. The oracle measures headroom and is never presented as a deployable baseline. Metrics follow the causal order of use: maximum diagnosed-lock violation; predicted contact-manifold proxy and realized minimum distance; physical contact; contact-candidate response RMSE; Spearman/Kendall and top-1 regret; and realized selected-candidate terminal error/success. The last stage is open-loop candidate selection, not closed-loop MPC.

D3 was excluded from gradient fitting and epoch selection, but earlier versions of the project inspected D3 diagnostics. We therefore call it held out from fitting, not untouched confirmation. Historical simplified-arm and calibrated GenkiArm spectra are retained below as disclosed development/failure evidence; they are not pooled with the strict primary table.

VI. Results

A. Strict six-stage fault-adaptation result

The primary comparison uses development domains D2/D4, seeds 7/17/27, 400 independent action groups per seed, 128 candidates per group, and a fixed 50-step horizon. Every method sees the same initial states, actions, realized candidate outcomes, and compute budget. D3 is excluded from this table because it was held out from gradient fitting but had been inspected during earlier development; it is therefore not represented as pristine confirmation data. The "outcome" below is the realized result of the selected open-loop candidate, not receding-horizon MPC.

The global residual reduces top-1 regret and terminal candidate error in 3/3 seeds. This is the strongest stable performance result, but it is deliberately not attributed to selective publication. Relative to the same-capacity global residual, selective IPWM changes mean regret by -2.07%, terminal error by -0.32%, success by -0.42 points, and Spearman by only $+0.00037$; moreover, full-state and selective publication give identical outputs in all three seeds. The selective-structure attribution gate is therefore No-Go.

The realized-cost oracle has mean terminal error 0.03758 m versus 0.04675 m for nominal selection, leaving 9.17 mm or 19.63% mean endpoint headroom (positive in 3/3 seeds). The oracle is privileged and non-deployable; it shows that candidate quality is not exhausted, while the global residual closes only a fraction of the available outcome gap.

We additionally registered a post-freeze D3 candidate-query test before generating its seed-91031 archive (200 groups,

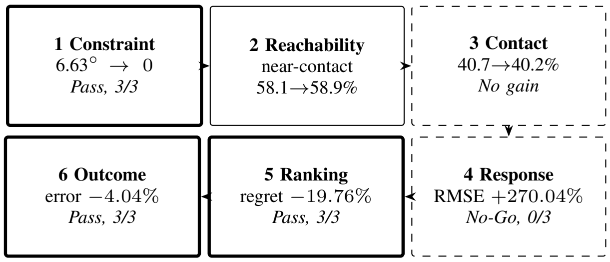

Fig. 2. Six-stage evidence chain for the same-capacity global residual versus the nominal world model. Solid boxes pass a frozen direction/constraint gate; dashed boxes expose a failed transfer. "Outcome" is the realized selected open-loop candidate, not receding-horizon MPC.

TABLE III
ANALYTIC-PROJECTION REMOVAL ABLATION OVER THREE SEEDS.

Constraint metric	No projection	Projection	Seeds
Max position drift (rad)	0.077–0.153	0	3/3
Mean position drift (deg)	6.63	0	3/3
Max velocity violation (rad/s)	0.384–0.660	0	3/3

128 candidates, 50 steps). D3 had been inspected in historical exploration, so this is not presented as pristine domain confirmation. Global residual versus nominal reduces regret by only 9.77% (2/3 seeds) and terminal error by 2.00% (2/3), while success rises 2.17 points (3/3); response RMSE worsens 263.63% and Spearman changes by −0.0148. Selective IPWM again fails matched attribution: relative to global, regret changes by −3.82%, terminal error by −0.50%, and success by −0.83 points. Thus the fresh-query result confirms no large transferable task-performance advantage and bounds the D2/D4 development claim.

The response column exposes the central diagnostic result. Both decision-trained models choose better actions while their contact-candidate response RMSE becomes substantially worse. Consequently, neither one-step nor rollout RMSE can stand in for action-ranking evidence in fault-aware contact planning.

Table III isolates a different contribution. Removing projection permits large diagnosed-joint drift, whereas analytic projection sets both violations to exactly zero. Candidate regret and terminal outcome change little, so projection is claimed as a structural correctness guarantee, not as a success-rate improvement.

B. Decision-loss and publication ablations

With the same architecture and 40-epoch protocol, weight-10 counterfactual decision training versus weight zero improves top-1 regret by 5.98% and terminal error by 1.40% on average, but only in 2/3 seeds; it decreases success by 0.50 points and worsens response RMSE by 356.44%. Conversely, weight zero improves response RMSE by 24.84% and success by 1.67 points over nominal in 3/3 seeds. The paired objective therefore shifts the model along a prediction–decision tradeoff; it is not a uniformly improving component.

TABLE IV
HISTORICAL CALIBRATED-GENKIARM OBJECT-RMSE IMPROVEMENT (%) OVER THE MECHANISM-MATCHED CARRIER; RETAINED AS A NEGATIVE TRANSFER BOUNDARY.

Method	107	117	127	Mean
Routed selective	0.604	-0.784	0.718	0.179
Raw selective	1.420	-1.913	2.546	0.684

TABLE V
ISOLATION ABLATION ON THE POST-DESIGN QUERY SET. OBJECT GAIN IS THE MEAN OVER CHECKPOINT-SEED MEANS; FREE REGRESSION IS THE WORST CELL.

Method	Object gain	Free regress.	Positive free regress.
Carrier	0	0	0/27
Full-state IPWM	16.74%	6.20%	3/27
SI-IPWM	16.74%	0	0/27

C. Retrospective calibrated-GenkiArm boundary

The earlier 27/37/47 checkpoint matrix below was generated on the simplified `arm_push.xml` development model. It is retained to explain the origin of state isolation, but it is not the paper's calibrated-model performance evidence. The retrospective audit used `genkiarm_push.xml`, disjoint query seeds, and fresh training seeds 107/117/127.

For routed selective SI-IPWM, only 2/3 seeds are positive and the hierarchical paired-bootstrap 95% interval is [-0.745%, 0.797%]. The raw selective interval is [-1.870%, 2.502%]. Both fail the preregistered +5% mean, 4/5 positive-seed, and positive-lower-bound gates. Free-state regression and locked-coordinate violation are exactly zero in all three seeds. The retrospective decision is therefore performance No-Go and mechanism-only support. These rows are not pooled with the strict primary result.

D. Historical simplified-model diagnosis

E. The full-state intervention has a systematic failure

On seed 7, D3 mixed-unseen, H50, the original intervention has free-joint RMSE 0.808 versus 0.762 for the shared projected baseline, a 6.01% regression. The paired MSE-difference interval is [0.06885, 0.07538], and 93.3% of terminal windows regress. Object RMSE improves 36.49% and lock violation remains zero. The failure therefore exposes a real tradeoff: object correction feeds back into unaffected robot prediction even though feasibility is projected.

The post-design query set makes the repair comparison direct (Table V). Full-state and isolated methods have exactly the same object RMSE in every cell because the intervention keeps its private physical trajectory. Full-state rollout nevertheless regresses free-joint RMSE in 3/27 cells, with a 6.20% worst case. Isolation removes those changes without changing object prediction.

F. State isolation removes collateral regression

Table VI reports the post-design query-set cells. Object RMSE improves in 27/27 cells. The smallest gain is 1.58%

TABLE VI
OBJECT RMSE IMPROVEMENT (%) OVER THE MECHANISM-MATCHED CARRIER IN ACTIVE PHYSICAL-OOD ROUTES.

Seed	Condition	H10	H25	H50
27	high damping	7.05	3.06	7.58
27	composition	7.70	2.03	19.54
27	mixed unseen	9.08	3.26	3.43
37	high damping	7.31	21.64	37.41
37	composition	8.52	20.93	30.23
37	mixed unseen	10.39	21.00	1.58
47	high damping	8.57	34.68	33.00
47	composition	10.63	35.53	30.52
47	mixed unseen	12.53	38.86	25.95

TABLE VII
AGGREGATE PREDICTION AUDIT. "CHANGE" IS RELATIVE TO CARRIER.

Statistic	Result
Positive object cells	27/27
Object improvement range	1.58–38.86%
Mean over checkpoint-seed means	16.74%
Seed-bootstrap 95% interval	[6.97, 25.59]%
Maximum |free-joint change|	0
Maximum |pusher change|	0
Maximum lock violation RMSE	0

(checkpoint seed 37, mixed unseen, H50), and the largest is 38.86%. Mean improvement over the three checkpoint-seed means is 16.74%, with descriptive bootstrap interval [6.97%, 25.59%]. The original seed-specific query sets independently retain 27/27 positive cells, 1.34–39.97% range, and 17.04% mean. Across all cells, free-joint and pusher changes relative to the carrier are numerically exactly zero, as predicted by Eq. (5); lock violation RMSE is zero.

The four low-context conditions route to the carrier and therefore tie it exactly. This is intentionally conservative: an earlier topology-only router improved severe physical shifts but degraded nominal-like H25/H50 cells by as much as 32.73%. A K25 support-validation selector was also rejected because the short active probe contained insufficient object contact and would reject every useful intervention. These negative controls motivate physical-context routing but do not establish that norm thresholding is optimal.

G. Routing margins and query replication

Table VIII reports context-norm ranges across the three checkpoints on the original physics spectrum. Nominal, noisy-deadband, and weak-motor conditions have wide margins below 1.2. Delay is closer (maximum 1.12), and mixed unseen is the closest active condition (minimum 1.26). Thus the observed decisions are stable on these checkpoints, although the 0.06 mixed-unseen margin is too small to call the router calibrated under arbitrary shift.

The post-design query set changes both calibration and evaluation trajectories while leaving weights, K, scale, threshold, and domains frozen. Its 16.74% mean is close to the original 17.04%; both have 27/27 positive active cells. This replication argues against memorization of one action/initialization set. It does not test a new object geometry, simulator, or training seed, and is labelled accordingly throughout.

TABLE VIII
CONTEXT NORM RANGE AND FROZEN ROUTE ACROSS CHECKPOINT SEEDS.

Condition	Minimum	Maximum	Route
Nominal	0.49	0.52	carrier
Noisy deadband	0.48	0.52	carrier
Weak motor	0.77	0.83	carrier
Delay 1	1.07	1.12	carrier
High damping	1.33	1.54	intervention
Composition	1.45	1.50	intervention
Mixed unseen	1.26	1.45	intervention

TABLE IX
D3 HIGH-DAMPING MEAN TERMINAL DISTANCE (MM) OVER THREE HELD-OUT TARGETS. LOWER IS BETTER.

Seed	Nominal IK	Carrier MPC	SI-IPWM MPC
27	16.99	13.75	19.45
37	16.99	18.77	15.67
47	16.99	34.25	39.11

H. Prediction gain does not imply control gain

We evaluated a frozen guarded CEM planner using 60 approach and 90 push steps, horizon 3, eight candidates, one CEM update, and a fixed 0.85 action-deviation guard. The prior 50 mm success tolerance is invalid for this target split: targets start only 30–40 mm from the object, so zero-contact episodes count as success. We instead report terminal distance, contact, and displacement.

All audited episodes make contact, but SI-IPWM improves on carrier MPC for only one of three seeds. The closed-loop improvement claim is therefore No-Go. This result is consistent with the method's guarantee: isolation controls which state is published, not how a finite-sample optimizer converts prediction differences into actions.

I. Reproducibility

Frozen YAML files define domain splits, context scaling, adapters, and routing. Evaluation artifacts include aggregate and per-window squared errors. The original selective audit contains 5,040 raw rows per seed, and the post-design query audit contains 2,160 per seed; a separate script regenerates Table VII and its bootstrap interval. Unit tests verify exact projection, carrier publication, and preservation of private intervention feedback with a robot-coupled toy model. Failed gates and protocol corrections are retained in the repository rather than overwritten.

J. Threats to validity

Post-hoc structure. State isolation was invented after the original free-state failure; only query seed 57 is post-design. *Dependence.* The 27 cells share three checkpoints and three domains, so they are not treated as 27 independent replicates. *Baseline scope.* Carrier matching isolates the intervention

TABLE X
RETAINED FAILURES AND THEIR CONSEQUENCE.

Attempt	Observed failure	Decision
Hard mask only	contact not established	separate feasibility/contact
No analytic projection	mean lock drift 6.63 deg	retain hard projection
Decision weight 10	response RMSE -356.44% vs. weight 0	report tradeoff
Global residual	regret -19.76%, 3/3	control-signal Pass
Selective vs. global	aggregate regret -2.07%	attribution No-Go
Direct guarded MPC	improves 1/3 historical seeds	MPC No-Go

effect, but comparisons to AdaptiGraph or PIN-WM are conceptual rather than reproduced. *Metric scope.* RMSE weights all object coordinates uniformly and does not imply useful action ranking. The No-Go MPC result makes this boundary visible. *Router scope.* Context norm is an engineering support heuristic, not a calibrated probability of harm. *Simulation scope.* The shared variable-node transition Gate improves both GenkiArm and Panda in 2/3 seeds, but the downstream object/contact Gate is No-Go in 0/3 seeds. A no-weld scripted Panda grasp succeeds in 5/5 small pose perturbations, and both fixed eye-to-hand views retain object visibility under the frozen extrinsic perturbations; these are task/observability feasibility, not learned Grasp, RGB prediction, or contact-transfer evidence.

VII. Failure Ledger and Decision Rules

The development history is part of the evidence because each accepted component corresponds to a measured failure (Table X). We use “Pass” only for a claim whose frozen metric succeeds; a method can pass prediction while remaining No-Go for control. This prevents the exact lock constraint, which is easy to satisfy by projection, from masking degradation of free coordinates or task outcomes.

The claim acceptance rule is consequently stage-specific. Locked coordinates must have zero violation; reachability and physical contact are reported before response error; ranking requires paired candidates; and task benefit requires realized outcomes under the same selection budget. The current artifact passes the hard-constraint and nominal-to-adapted regret gates. It fails selective structure attribution, response-RMSE improvement, and receding-horizon MPC.

A. Artifact-to-claim mapping

Three independently checkable artifacts underlie the paper: raw trajectory windows support object statistics, model outputs support isolation equality, and environment logs support contact and terminal-distance checks. Aggregate tables are regenerated rather than hand selected. The code path accepts an explicit query seed so that checkpoint initialization and evaluation queries can be varied separately. Hardware logs, photographs, and measurements are not inputs to any table in this submission.

VIII. Evaluation and Audit Details

A. Primary candidate groups and aggregation

Each seed contributes 400 groups and each group contains 128 candidates sharing one initial state, goal, and diagnosis. Spearman and Kendall correlations are computed within groups and averaged over groups. Top-1 regret is the realized cost of the model-selected candidate minus the minimum realized cost in that same group. Terminal error and success are evaluated on the selected physical candidate, never on its prediction. Seed-level relative changes are computed first and then averaged; the three seeds, rather than 153,600 candidates, are the independent units used for direction counts. Continuous minimum geometry distance and simulator contact are stored separately so that near-contact is not relabeled as contact.

TABLE XI
FROZEN CLAIM GATES AND CURRENT OUTCOME.

Claim	Gate	Outcome
Lock feasibility	position/velocity violation $= 0$	Pass (3/3)
Action ranking	regret reduction, consistent direction	Pass (3/3)
Selected outcome	terminal-error reduction, consistent	Pass (3/3)
Response prediction	RMSE improvement	No-Go (0/3)
Selective attribution	beat global control consistently	No-Go
Receding-horizon MPC	consistent seed direction	No-Go (1/3)

B. Window construction and aggregation

Each test trajectory contains 150 actions and 151 states. For horizon H, autonomous prediction starts at indices separated by H and rolls for exactly H actions; only the terminal prediction enters that window’s statistic. Squared error is computed per trajectory/window, averaged, and square-rooted to obtain RMSE. Free-joint error masks position and velocity of the diagnosed coordinate. Object error averages four planar coordinates. Pusher error applies the same forward kinematics to predicted and true joint positions.

The original spectrum audit stores 5,040 per-window/method rows per checkpoint. The post-design audit restricts generation to three active regimes and, after adding the full-state ablation, stores 2,880 rows per checkpoint. Carrier, full-state, isolated, and routed predictions use identical initial states and actions. Exact object equality between full-state and isolated rollouts is checked cell by cell rather than inferred from their aggregate.

C. Uncertainty and multiplicity

The bootstrap treats the mean across nine active cells within a checkpoint as one observation and resamples three checkpoint means. Correlated horizons and domains are therefore not presented as 27 independent seeds. With only three observations, the interval is a stability summary, not precise population inference. We consequently report every cell and the positive-cell count; no null-hypothesis threshold accepts the main claim.

The seed-7 diagnostic interval has a different purpose: it bootstraps paired terminal-window MSE differences on one prespecified failure cell. It shows the free-state regression is systematic within that query set, not that it generalizes to a population of training seeds. Keeping the two analyses separate prevents trajectory count from inflating the effective number of model initializations.

D. Controller audit protocol

All controller methods share analytic approach/push references, damage mask, action limits, initial block position, and held-out targets. Guarded CEM uses horizon 3, eight candidates, one update, and falls back to nominal IK when the proposal differs by more than 0.85 in Euclidean norm. Sixty approach steps precede 90 push steps; the earlier 30+40 protocol produced zero-contact episodes and remains only as a failed smoke test. Contact count and block displacement are required sanity metrics.

We do not report the old 50 mm success indicator because initial target distance is at most 40 mm. A valid future success metric must be stricter than every initial distance and require contact and minimum displacement. The paper uses terminal distance for the completed controller matrix and refuses a binary success claim. This correction applies to all methods simultaneously.

E. Reproduction boundary

The evaluator records checkpoint and query seeds separately. Reproducing the main table requires the three strict checkpoints, V2 candidate archives for seeds 7/17/27, and the fixed 400-by-128-by-50 evaluator. The formal checkpoint loader rejects missing, unexpected, or shape-mismatched tensors. Separate summarizers regenerate the strict primary result, decision-loss comparison, same-capacity global control, and projection-removal ablation from JSON without loading weights. Tests cover pairing, held-out-lock exclusion, the soft-regret objective, exact projection, and private/published state recurrence.

Training is not yet packaged as a single archival container, and checkpoints depend on staged predecessor runs. This limits reproduction convenience even though evaluation is scripted. A camera-ready artifact should add immutable hashes and one top-level command; we do not claim that packaging is complete.

IX. Limitations and Conclusion

This study is simulation-only and contains no real-robot evidence. The planned hardware protocol uses two fixed eye-to-hand cameras, one overhead and one horizontal; neither source is eye-in-hand. The present results must not be interpreted as sim-to-real validation.

The hardware protocol is split before collection. Level A executes one frozen, manually validated position trajectory per intact/D2/D3 condition (at least 10 valid trials each) and can support only lock, reach, contact, and Push feasibility. Level B would compare learned selectors, but is unauthorized until a measured bridge maps MuJoCo generalized-force actions to the arm's servo goal-position interface and freezes a common candidate library. Schedule and library hashes, all aborts, two videos, and joint logs are retained. Thus a fixed-trajectory result cannot be relabeled as learned recovery.

Only single locks and planar pushing are evaluated. Diagnosis must supply the correct lock identity and angle. The primary performance table uses D2/D4 development evidence rather than a pristine confirmation domain, and realized open-loop candidate outcomes are not receding-horizon MPC. Panda object/contact transfer is No-Go in 0/3 seeds; its scripted 5/5 grasp result and dual fixed eye-to-hand visibility audit establish feasibility only. No real-robot data are reported in the current manuscript version.

The strongest supported conclusions are deliberately narrow. Analytic projection changes a learned roll-out's 4.42–8.75 degree maximum lock drift to exactly zero in 3/3 seeds. Fault adaptation reduces top-1 candidate regret by 19.76% against nominal in 3/3 seeds, but response RMSE degrades severely and a same-capacity global residual matches or exceeds selective IPWM on aggregate. Thus the evidence supports hard constraint satisfaction, prediction–decision decoupling, and a reusable six-stage diagnostic; it rejects the claim that selective publication causes the current control-related gain. Historical simplified-model tables remain disclosed development evidence, not confirmation.

References

[1] A. Cully, J. Clune, D. Tarapore, and J.-B. Mouret, "Robots that can adapt like animals," *Nature*, vol. 521, pp. 503–507, 2015.

[2] F. Yang, C. Yang, D. Guo, H. Liu, and F. Sun, "Fault-aware robust control via adversarial reinforcement learning," in *IEEE CYBER*, 2021.

[3] A. Kumar, Z. Fu, D. Pathak, and J. Malik, "Rma: Rapid motor adaptation for legged robots," in *Robotics: Science and Systems*, 2021.

[4] A. Allevato, E. Short, M. Pryor, and A. Thomaz, "Tunenet: One-shot residual tuning for system identification and sim-to-real robot task transfer," in *Conference on Robot Learning*, 2020.

[5] C. Finn and S. Levine, "Deep visual foresight for planning robot motion," in *IEEE ICRA*, 2017.

[6] M. Bauza, F. R. Hogan, and A. Rodriguez, "A data-efficient approach to precise and controlled pushing," in *Proceedings of the 2nd Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 87, 2018, pp. 336–345.

[7] S. Kim, B. Lim, Y. Lee, and F. Park, "SE(2)-equivariant pushing dynamics models for tabletop object manipulations," in *Conference on Robot Learning*, 2023.

[8] S. Richards, N. Azizan, J.-J. Slotine, and M. Pavone, "Adaptive-control-oriented meta-learning for nonlinear systems," in *Robotics: Science and Systems*, 2021.

[9] A. Nagabandi, G. Kahn, R. Fearing, and S. Levine, "Neural network dynamics for model-based deep reinforcement learning with model-free fine-tuning," in *IEEE ICRA*, 2018.

[10] H. Gao, Y. Ouyang, and M. Tomizuka, "Online learning in planar pushing with combined prediction model," *arXiv:1910.08181*, 2019.

[11] K. Zhang, B. Li, K. Hauser, and Y. Li, "Adaptigraph: Material-adaptive graph-based neural dynamics for robotic manipulation," in *Robotics: Science and Systems*, 2024.

[12] W. Li, H. Zhao, Z. Yu, Y. Du, Q. Zou, R. Hu, and K. Xu, "Pin-wm: Learning physics-informed world models for non-prehensile manipulation," in *Robotics: Science and Systems*, 2025.

[13] Z. Zhong, S. Golestaneh, and C. Chamzas, "Activepusher: Active learning and planning with residual physics for nonprehensile manipulation," in *IEEE International Conference on Robotics and Automation*, 2026.

[14] P. Battaglia, R. Pascanu, M. Lai, D. Rezende, and K. Kavukcuoglu, "Interaction networks for learning about objects, relations and physics," in *NeurIPS*, 2016.

[15] A. Sanchez-Gonzalez, J. Godwin, T. Pfaff, R. Ying, J. Leskovec, and P. Battaglia, "Learning to simulate complex physics with graph networks," in *ICML*, 2020.